

Sports, politics, and social media: A Human-AI collaborative analysis of consumer reactions to Trump's break 50 appearance

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ABSTRACT

This study investigates consumer engagement on social media platforms in response to Donald Trump's participation in Bryson DeChambeau's Break 50 golf series, focusing on audience interpretation of politically infused sports entertainment. Drawing on framing theory and stereotype content model (SCM), we systematically examined YouTube and Instagram comments through an innovative human-AI collaboration pipeline integrating BERTopic, reasoning-enhanced large language models (LLMs), LLM fine-tuning, and embedding-based semantic evaluations. Our analysis identified three distinct generic frames, entertainment, politics/controversy, and athletics, each eliciting unique warmth and competence evaluations. Entertainment frames consistently occupied SCM's admiration quadrant, whereas political discourse more often aligned with contempt. However, several issue-specific frames deviated from traditional SCM profiles, revealing nuanced audience perceptions such as warm-but-incompetent and cold-but-competent stereotypes. Cross-platform comparisons uncovered evaluative reversals, indicating that perceptions of Trump shifted with platform norms and audience composition. Theoretically, the findings extend framing theory by illustrating how audiences actively co-construct meaning in politically salient entertainment contexts, while reaffirming SCM's utility in analyzing spontaneous digital discourse. Methodologically, we demonstrate a scalable, replicable human-AI workflow for large-scale, theory-driven social media analysis. Practically, the results offer insights for content creators, sports marketers, and political strategists seeking to navigate the reputational opportunities and risks of aligning with polarizing figures in cross-platform media environments.

1. Introduction

Social media has fundamentally reshaped how athletes and public figures engage with their audiences, enabling direct interaction and dialogue through user-generated content (UGC), a marked shift away from traditional, one-way media communication such as televised broadcasts (Filo et al., 2015; Peng, 2021). This shift is especially evident in sports like golf, historically reliant on conventional media but now embracing social media platforms for audience interaction (PGA, 2025). Bryson DeChambeau's popular YouTube series Break 50, exemplifies this digital transformation by documenting his attempts to complete golf rounds in fewer than 50 strokes, notably including an episode featuring the then former U.S. President Donald Trump during his 2024 presidential campaign. The explicit integration of sports performance with

political identity in this episode creates a unique opportunity to explore precisely how social media users frame their perceptions and judgments of politically charged sports content, offering insights that extend beyond viewership metrics.

Even so, existing scholarship has rarely examined how users themselves construct meaning when sports content is intertwined with political figures (Larkin, 2025). Prior work in political communication, digital marketing, and public relations has documented how audiences respond to political actors (Bennett et al., 2019; Peng, 2021; Qian et al., 2024) and how they engage with sports narratives (A. P. Kirilenko and Stepchenkova, 2017; T. Y. Qian et al., 2024; Xu et al., 2024), yet these studies have developed largely in parallel. As a result, little is known about the interpretive processes that occur when these domains converge, particularly in highly visible, politically charged sports

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moments. This absence is striking given that such intersections could polarize public discourse, influence brand and athlete reputations, and shape the mediated personas of both political and sports figures (Larkin, 2025; Peng, 2021). Addressing this gap is therefore essential for understanding how social media users selectively emphasize aspects of hybrid sports-political content in ways that reflect personal beliefs, ideological alignments, and community norms. Accordingly, our research seeks to address the following research questions (RQs):

RQ1: What specific themes emerge in social media users' reactions to politically infused sports content, specifically Trump's appearance on Break 50?

RQ2: How do social media users' perceptions of character attributes interact with the identified themes?

RQ3: How do these themes and associated evaluative judgements differ between platforms (YouTube vs. Instagram)?

To answer these RQs, we draw upon framing theory, emphasizing frames as reflections of individuals' cognitive perceptions expressed through social interactions. Framing refers to how people conceptualize an issue or event by selectively highlighting specific aspects based on their personal beliefs, assumptions, and expectations (De Vreese, 2005; Entman, 1993). Thus, examining frames in social media conversations reveals the underlying cognitive frameworks guiding audience responses. By analyzing UGC in the form of audience comments, we identify and interpret these audience-driven frames, uncovering how perceptions around politically infused sports content are collectively constructed and communicated. Complementing framing theory, we integrate stereotype content model (SCM), which posits that individuals typically evaluate others along two primary dimensions: warmth (trustworthiness, friendliness) and competence (ability, effectiveness) (Fiske et al., 2007). This conceptual model allows us to capture subtle dimensions of audience evaluations, providing a nuanced understanding of how social media users frame their perceptions of public figures like Trump and DeChambeau, especially in contexts that blend athletic performance with political identity.

Faced with the sheer volume and diversity of UGC, we employ natural language processing (NLP) techniques to analyze audience comments from YouTube and Instagram concerning the Trump episode of Break 50 (Fig. 1). First, we apply BERTopic, an emerging topic modeling technique, to systematically cluster semantically similar audience comments, thereby identifying dominant interpretive themes (Grootendorst, 2022). These identified themes subsequently guide our human-AI collaboration pipeline leveraging large language models

(LLMs). Specifically, we utilize OpenAI's o4-mini as an analytical assistant and fine-tuning GPT-4o mini to conduct a hierarchical analysis of both generic and issue-specific frames (De Freitas et al., 2025; De Vreese, 2005; OpenAI, 2023, 2025c, 2025b). Additionally, we operationalize the SCM's dimensions of warmth and competence using the single-category word embedding association test (S-WEAT), quantifying how strongly these dimensions emerge within user-generated frames through semantic embeddings and cosine similarity measures (Charlesworth et al., 2023; Fiske et al., 2002; Kurdi et al., 2019).

The present study advances both scholarly understanding and practical applications. Theoretically, we expand the application of framing theory by exploring how user-generated frames in social media reflect audience cognition and interpretation in politically infused sports scenarios. Moreover, by integrating SCM, we offer fine-grained insights into the semantic dimensions shaping these frames. Methodologically, our combination of BERTopic and LLMs demonstrates an innovative approach for analyzing large-scale, unstructured social media datasets, illustrating the practicality and effectiveness of advanced NLP techniques driven by a human-AI collaborative workflow. Empirically, this research produces valuable insights for practitioners across multiple domains. Marketers, content creators, and political strategists might gain clarity on how audiences frame interactions between sports, entertainment, and politics. Specifically, analyzing reactions to the Trump-DeChambeau collaboration sheds light on how online users respond to explicit intersections between sports performance and political personas, offering practical insights into effective audience engagement strategies in politically sensitive sports marketing.

The manuscript proceeds as follows: First, we review existing literature on social media framing and explain the theoretical and analytical frameworks. Next, we outline our methodological approach, detailing data collection, NLP methods, and analytical strategies. We then present and interpret the results, followed by a discussion of theoretical implications, methodological implications, practical contributions, study limitations, and avenues for future research.

2. Literature review

2.1. Meaning-making and framing across social media platforms

Framing theory, initially posited by Entman (1993), fundamentally addresses how information is selectively emphasized in communication to shape audience interpretation. Essentially, framing involves selecting particular aspects of perceived reality and highlighting them to define issues, attribute causes, make moral evaluations, and recommend

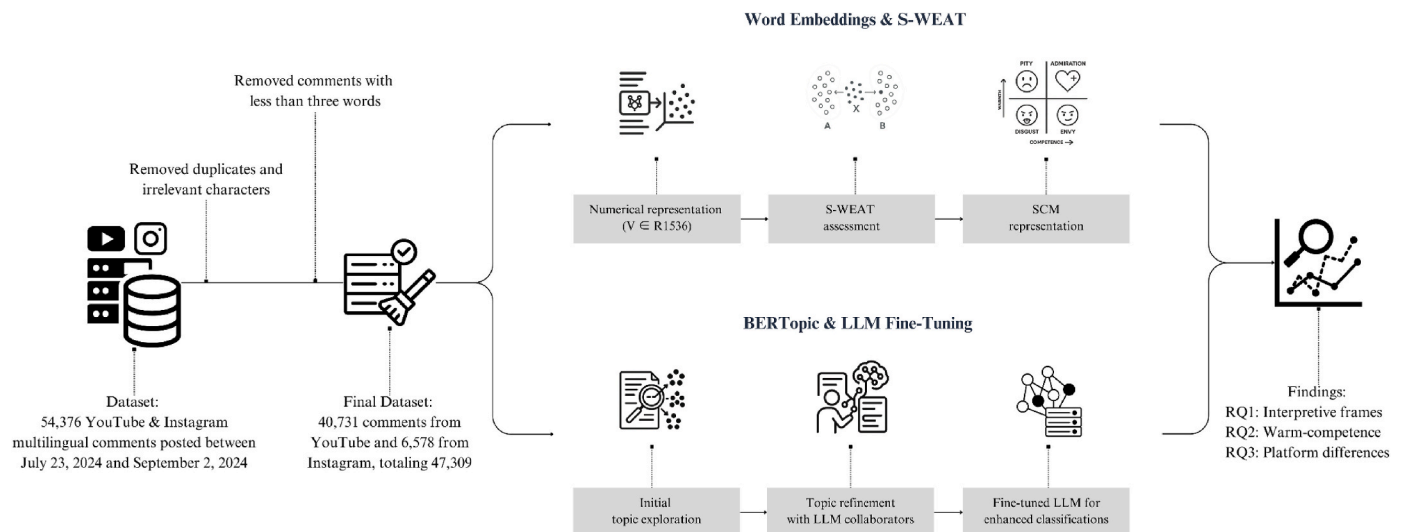


Fig. 1. Research roadmap.

remedies. Building on this foundation, De Vreese (2005) conceptualizes framing as a dynamic process involving frame-building, frame-setting, and subsequent effects. Traditionally, framing studies have predominantly examined how mass media sources, such as news organizations, shape public discourse through editorial choices (Billings and Eastman, 2003; Guo et al., 2021; Jiang et al., 2024; Tan and Cheng, 2024). However, the proliferation of social media platforms has fundamentally transformed this dynamic, empowering users themselves as active participants in the framing process (Dwivedi et al., 2021). The integration of framing theory into the analysis of UGC represents a shift from viewing audiences as passive receivers to recognizing their role as active, interpretive co-creators.

Indeed, social media platforms have uniquely enabled audiences to construct and publicly share their own interpretive frames in response to events, transcending traditional media's unilateral influence (Chaudhuri et al., 2025; Dwivedi et al., 2021; Xu et al., 2024). Unlike traditional media frames, UGC frames are expressions of individual audience members' cognitive schemas and interpretations, externalized through their social media interactions (Ocal and Crowston, 2024). Arguably, UGC may reveal both individual cognitive frameworks and collective meaning-making processes shaped by shared beliefs, assumptions, and expectations, making social media platforms fertile ground for investigating how individual and group interpretations intersect, diverge, or evolve (Entman, 1993; Ocal and Crowston, 2024; Xu et al., 2024). This is particularly pertinent in politically charged contexts. When politics intersects with popular sports media, audience framing may become increasingly nuanced, as individuals negotiate multiple identities and conflicting schemas (Bennett et al., 2019; Larkin, 2025; Wang et al., 2024).

Given these theoretical underpinnings, our study explicitly applies framing theory within the context of UGC, analyzing YouTube and Instagram comments related to Trump's appearance on DeChambeau's Break 50. Following De Vreese's (2005) typology, we distinguish between generic and issue-specific frames to provide a systematic basis for analysis. Generic frames transcend specific thematic boundaries, enabling the identification of broader interpretive patterns. Referencing prior research in news media framing (A. Arora et al., 2025; De Vreese, 2005), we adapt generic frames including entertainment, politics/controversy, and athletics to capture general interpretive orientations. Issue-specific frames, in contrast, highlight how particular sub-themes are emphasized within discourse. While generic frames offer literature-based expectations (e.g., entertainment often cues perceptions of friendliness, charisma, and likability, leading to favorable perceptual evaluations) (Kervyn et al., 2022), issue-specific manifestations can vary considerably depending on context and audience (A. Arora et al., 2025). This variability makes the integration of issue-specific frames especially important in such a cross-domain collaboration, where the existence of multiple interpretations of the same content may elicit admiration from one audience segment and skepticism or contempt from another. As such, examining both frame levels balances conceptual grounding with sensitivity to the complexity of real-world online discourse: generic frames capture the stable interpretive logics established in prior research, while issue-specific frames highlight how those logics are inflected, challenged, or even contradicted by contextual dynamics.

This distinction becomes even more critical in cross-platform settings. Audience interpretations are shaped not only by the content of frames but also by the affordances and engagement cultures of different platforms. Prior multi-platform research has shown that demographic profiles, motivational differences, and interactional norms can significantly influence how frames are constructed and received (A. Kirilenko et al., 2024; S. Qian et al., 2024). For instance, Instagram's visual and relational orientation tends to foreground aesthetic, interpersonal, and identity cues, whereas YouTube's long-form, persistent comment spaces are more conducive to extended, narrative-driven exchanges (Peng, 2021). By distinguishing between generic and issue-specific frames within these different platform contexts, our study develops a

theoretically grounded yet flexible approach that captures both theoretical associations and context-dependent complexities.

2.2. Evaluative judgments and perceived traits in UGC

Alongside distinct frames, audience reactions frequently entail evaluative judgments about public figures. The stereotype content model (SCM) offers a robust framework for systematically understanding these evaluations, suggesting that social perceptions of individuals and groups primarily organize around two dimensions: warmth and competence (Fiske et al., 2002). Warmth encompasses attributes like friendliness, trustworthiness, and moral intention, while competence relates to perceived ability, effectiveness, and skill. These two dimensions create four distinct quadrants of social perception (Fiske et al., 2007). High warmth and high competence (Quadrant I) characterize admired groups, evoking pride and admiration. High warmth and low competence (Quadrant II) describe pitied groups, eliciting sympathy and paternalistic attitudes. Low warmth and low competence (Quadrant III) encompass contemptuous groups, provoking disgust and active harm. Finally, low warmth and high competence (Quadrant IV) define envied groups, generating respect mixed with resentment.

SCM initially appeared in social psychology research examining intergroup relations but has since expanded across social sciences (Bennett et al., 2019; Durante et al., 2010; Funk, 1997; Kervyn et al., 2012, 2022, 2022; Liu et al., 2024). Crucially, SCM has proven effective for textual analysis, enabling researchers to systematically categorize audience-generated content based on warmth-competence assessments (Liu et al., 2024). For instance, Kurdi et al. (2019) and Lomas et al. (2025) have demonstrated the utility of SCM-related lexicons in coding open-ended textual responses, allowing quantification and comparison of evaluative judgments within large text corpora. Charlesworth et al. (2023) recently demonstrated the enduring relevance of SCM in analyzing historical language, showing that descriptors used to characterize social groups often change superficially but consistently reflect underlying warmth and competence judgments. This indicates that SCM dimensions are foundational cognitive tools for interpreting social stimuli, persisting across cultures and historical contexts. Such robustness highlights the theory's suitability for analyzing UGC, where users make judgments about public figures' character traits, capturing nuanced audience perceptions often obscured by broader sentiment analysis.

Therefore, to complement framing theory, which identifies the interpretive schemas embedded in user-generated discourse, our study employs SCM as a complementary theoretical framework to analyze audience evaluations of Trump within the sports media context. Specially, SCM offers an evaluative lens that captures how UGC frames activate warmth and competence judgments. By coding user comments for warmth- and competence-related terms (Kurdi et al., 2019), we systematically quantify the evaluative landscape surrounding this politically charged sports content collaboration. This theoretical integration allows for an augmented assessment that moves beyond mere descriptive or valence-only approaches (e.g., simplistic approval or disapproval), clarifying audience positions in terms of socially meaningful trait attributions and the frames through which those traits are activated. By embedding SCM within a framing analysis, our study offers a theory-guided and transparent design that not only enriches conceptual interpretation but also aligns with reproducible practices, thereby contributing both to theoretical development and to the methodological advancement of computational social science (Baier et al., 2025; Basu et al., 2024; Berente et al., 2019; T. Y. Qian et al., 2025).

2.3. Computational methods for UGC analysis

Given the massive scale and complexity of social media content, traditional qualitative analyses are insufficient for comprehensive and efficient insight generation. NLP techniques have emerged as

indispensable tools for systematic analysis of large-scale UGC datasets, enabling precise identification of themes, frames, and sentiment nuances (Basu et al., 2024; Berente et al., 2019; Pandey and Singh, 2025; Perazzoli et al., 2022; Silayach et al., 2025; Ullah et al., 2025; Zha et al., 2023).

Topic modeling is a central NLP technique for uncovering prevalent themes in unstructured textual data. BERTopic represents a cutting-edge, embedding-based approach, surpassing traditional methods like Latent Dirichlet Allocation (LDA) by leveraging transformer models such as BERT for semantic context awareness (Grootendorst, 2022). This capability allows BERTopic to effectively cluster short, contextually rich texts, such as social media comments, into coherent thematic groups, facilitating the identification of dominant audience frames without predefined categories (Egger and Yu, 2022; Jiang et al., 2024; A. P. Kirilenko and Stepchenkova, 2025; Son and Park, 2025).

Complementing BERTopic, word embedding techniques markedly improve the granularity of content analysis by capturing nuanced semantic relationships between terms. Word embeddings transform textual data into vector representations capturing semantic relationships, thus enabling researchers to quantitatively assess nuanced linguistic patterns, including implicit evaluative judgments (Charlesworth et al., 2023; Egger, 2024). For instance, embedding-based methods allow systematic detection of SCM dimensions within text by measuring semantic proximity between comments and predefined warmth- or competence-related lexicons (Kurdi et al., 2019; Liu et al., 2024). This sophisticated approach provides novel insight into subtle audience evaluations that simple sentiment analysis might miss, critically enriching our analysis of politically infused sports content.

Additionally, LLMs such as OpenAI's GPT models have revolutionized qualitative analysis of textual data (N. Arora et al., 2024; T. Y. Qian et al., 2025; Yoganarasimhan and Iakovetskaia, 2024). These models offer profound contextual understanding, capable of interpreting idiomatic language, sarcasm, and subtle rhetorical framing cues (Baier et al., 2025; Collins et al., 2021; Li et al., 2025; Pandey and Singh, 2025; Rossi et al., 2024). As collaborators, LLMs can assist researchers by refining cluster interpretation and performing nuanced frame classification without extensive manual coding (A. P. Kirilenko and Stepchenkova, 2025; Zhao et al., 2025; Zhou et al., 2024). Such collaboration improves analytical efficiency and interpretative accuracy, essential for large-scale analyses like our study. Relatedly, fine-tuning enhances the precision and cost-effectiveness of LLMs by training pre-trained models on carefully curated datasets, often requiring only modest amounts of data, thereby optimizing their performance for specific analytical tasks (OpenAI, 2023). Recent research indicated that fine-tuned LLMs achieve substantially higher accuracy than general-purpose models, particularly in domain-specific or nuanced textual analysis tasks (Baier et al., 2025; De Freitas et al., 2025; T. Y. Qian et al., 2025).

Taken together, our study proposes a novel human-AI collaboration workflow to examine how audiences interpret and evaluate politically infused sports content in the digital sphere. Building on prior research that has employed NLP techniques in isolation, we integrate these analytical tools into a unified pipeline capable of both identifying latent interpretive frames and capturing implicit cognitive associations. This approach enables a dual-layered analysis of UGC, examining not only what themes emerge in audience discourse but also how such themes are evaluated along key social perception dimensions. By grounding these computational methods in framing theory and SCM, we offer a scalable yet theoretically robust framework for understanding contemporary meaning-making in politicized sport contexts and beyond.

3. Methods

3.1. Operationalizing framing and stereotype content through human-AI collaboration

To investigate how consumers construct meaning around politically

infused sports content, we developed an automated, scalable, and replicable human-AI collaborative pipeline that integrates topic modeling, LLM classification, and embedding-based semantic association techniques. In addressing RQ1, we first employed BERTopic to extract latent topics, followed by refinement with OpenAI's o4-mini (OpenAI, 2025c) to assign topics to generic frames, operationalized from framing theory and pertinent empirical research (A. Arora et al., 2025; De Vreese, 2005; Entman, 1993). To address the interpretive limitations of unsupervised models, particularly the misclassification of outliers and the lack of theoretically grounded hierarchical relationships (Egger, 2024; Jiang et al., 2024; A. P. Kirilenko and Stepchenkova, 2025; T. Y. Qian et al., 2024; Son and Park, 2025), we introduced a fine-tuned GPT-4o-mini classifier (OpenAI, 2025b) trained on curated and validated datasets, assigning topics to issue-specific frames. Fine-tuning increased classification stability across runs and improved agreement with human-coded benchmarks, ensuring that frame categorization was both theoretically consistent and methodologically transparent.

To examine the evaluative dimensions of these frames (RQ2), we operationalized SCM constructs of warmth and competence using S-WEAT (Kurdi et al., 2019), anchored in the BAAI/bge-m3 embedding model (Chen et al., 2024). This approach quantifies implicit associations between frame-related content and theoretically derived warmth and competence term sets in high-dimensional vector space. Unlike topic prevalence or simple co-occurrence methods (e.g., Basu et al., 2024), S-WEAT captures perceptual nuance by directly measuring association strength. Lastly, we implemented the pipeline to both YouTube and Instagram datasets (RQ3), performing a systematic cross-platform comparison. This design not only reveals how framing and evaluative dimensions vary across platforms, but also addresses common limitations in prior UGC research, which often relies on single-platform, one-off, or non-replicable analyses. By uniting complementary theories within a robust computational workflow, our study advances both the methodological rigor and theoretical depth of politically salient sports discourse analysis.

3.2. Data collection and preprocessing

As shown in Fig. 1, we collected comments from the Break 50 video featuring Donald Trump, published on the verified @brysondechambeau YouTube channel, and the corresponding posts on the verified @brysondechambeau Instagram account. These platforms were selected due to their reach, UGC focus, and prominence in sports-related discourse (Filo et al., 2015; PGA, 2025). Comments from YouTube were retrieved using Google's YouTube Data API, while Instagram comments were obtained via Meta's Graph API, both in accordance with platform-specific data access protocols.

The data collection period spanned from July 23, 2024, to September 2, 2024, covering the peak timeframe during which the video and related Instagram posts received extensive audience interaction. A total of 54,376 comments were extracted. To ensure the accuracy, relevance, and coherence of our data, we performed a rigorous cleaning and preprocessing procedure. This included removing duplicate comments, as repetition could bias our topic modeling and framing analyses. We also stripped out URLs, numbers, user mentions, punctuation, special characters, and emojis, following standard preprocessing protocols established in previous UGC and social media research (Berger et al., 2020).

Additionally, we excluded comments shorter than three words. These short comments typically contained ambiguous statements or brief reactions (e.g., "Nice!", "Wow!", or "Great job!") that offered minimal interpretive value. Following these cleaning steps, our final preprocessed dataset contained 47,309 comments. Among these, 40,731 (86.10 %) were sourced from YouTube, while the remaining 6578 (13.90 %) originated from Instagram. Comments in our dataset varied considerably in length, ranging up to 1583 words. Most comments were concise, with a median length of 11 words and 75 % containing 20 words or fewer, though a minority engaged in extensive discussion. English

dominated the dataset, accounting for 93.76 % of comments, while the remaining 6.24 % comprised a diverse range of languages, primarily European and Asian.

3.3. Data analysis

3.3.1. Word embedding analysis with S-WEAT

To evaluate UGC in line with SCM, we adapted S-WEAT (Kurdi et al., 2019). This algorithm quantifies semantic associations by comparing each comment embedding against pairs of antonymous descriptor sets (Table 1). We computed average cosine similarities between comment embeddings and each descriptor set. Differences between opposing attributes were standardized within a range from -2.00 to $+2.00$ by dividing by the combined standard deviation of similarity scores. Positive standardized values indicated stronger associations with attributes like warmth and competence, whereas negative values implied closer associations with contrasting attributes.

$$s(X, A, B) = \frac{\text{mean}_{a \in A} s(a, X) - \text{mean}_{b \in B} s(b, X)}{\text{std_dev}_{w \in A \cup B} s(w, X)}$$

Where:

$$s(a, X) = \cos(\vec{a}, \vec{x})$$

Let $s(X, A, B)$ be defined over sets A and B representing dichotomous attribute word vectors, with X containing tweet word vectors; $s(a, X)$ quantifies cosine similarity between vectors, while $\text{std_dev}_{w \in A \cup B} s(w, X)$ normalizes the S-WEAT statistic.

For generating word embeddings, we employed OpenAI's text-embedding-3-small model (OpenAI, 2025d). This model converts text into 1536-dimensional numerical representations ($V \in \mathbb{R}^{1536}$), producing compact yet contextually accurate embeddings optimized for capturing nuanced semantic distinctions, ideal for large-scale, multilingual data. Recognizing potential variability, bootstrap resampling procedures were performed, repeatedly sampling 90.00 % of the dataset across 50 iterations to compute reliable average similarity metrics.

3.3.2. Integrative topic modeling with BERTopic

For BERTopic analysis, we first transformed comments into semantic embeddings using the BAAI/bge-m3 embedding model (Chen et al., 2024) via Hugging Face's Sentence Transformers library. This XLM-Roberta-based model converts text into 1536-dimensional numerical vectors ($V \in \mathbb{R}^{1024}$) and can process multilingual sequences up to 8194 tokens, making it particularly well-suited for our linguistically diverse dataset. Moreover, bge-m3 outperforms commonly used sentence-transformer baselines, especially on semantic textual similarity tasks, thereby ensuring greater topic coherence.

Subsequent embedding dimensionality reduction was performed via uniform manifold approximation and projection (UMAP) with a fixed random seed (42) for replicability. To identify emergent thematic structures, we clustered the document embeddings using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). To optimize the granularity of the resulting clusters, key parameters were iteratively tuned from a minimum of 25–100 in

increments of five. This careful tuning, combined with the algorithm's ability to handle noisy social media data, distinguishes our clustering approach from alternatives like K-means, which might introduce interpretive challenges by forcing every document into a predefined cluster. This forced categorization could obscure authentic patterns and hinder the goal of generating theoretically grounded insights from computational aggregations (Berente et al., 2019). The final stage used class-based term frequency-inverse document frequency (c-TF-IDF), enhanced by KeyBERT and Maximal Marginal Relevance (MMR), to identify representative keywords and documents. Initial topic labels were further refined using GPT-4.1 to enhance clarity and thematic coherence (OpenAI, 2025a).

Upon review of these refined labels, we identified considerable semantic overlap among several topics, suggesting potential for consolidation. To avoid obscuring meaningful distinctions during this process, we implemented hierarchical topic modeling as a complementary analysis (Grootendorst, 2022; Hu et al., 2025). It enabled systematic assessment of inter-topic relationships, providing proximity metrics on which topics should be merged to improve overall coherence while preserving distinct thematic elements. Notably, BERTopic identifies outliers (labeled as “-1”) that are semantically distant from main topic clusters. Although typically discarded in prior research or handled with simplistic algorithms such as c-TF-IDF representations for reassignment (Egger, 2024; A. P. Kirilenko and Stepchenkova, 2025; T. Y. Qian et al., 2024; Son and Park, 2025), we recognized not only the informational richness within these outliers but also their inherent complexity, necessitating improved processing methods. Thus, we proposed a structured pipeline integrating LLMs to systematically reassign these outliers, leveraging LLMs' contextual interpretive prowess to refine outlier classification and enhance topic interpretability (Berente et al., 2019).

3.3.3. Collaborative topic refinement with LLMs

Building upon the initial BERTopic analysis, we developed a collaboration pipeline integrating LLMs, specifically OpenAI's o4-mini, chosen for its optimized reasoning abilities (OpenAI, 2025c). The o4-mini model was provided with detailed BERTopic outputs, including hierarchical clustering results, topic labels refined by GPT-4.1, representative documents, and key terms. The goal was to leverage the model's structured semantic reasoning capability to classify and refine the identified topics into coherent interpretive frames.

As shown in Figs. 2 and 3, we designed a customized system prompt to guide our LLM collaboration process. This prompt instructed the model to apply established framing typologies, specifically generic frames and issue-specific frames to analyze our identified topics. The model was directed to categorize topics according to generic frames while providing justified recommendations for merging topics within issue-specific frames. To mitigate the inherent stochasticity of LLM outputs (Yoganarasimhan and Iakovetskaia, 2024), we executed this structured analytical pipeline ten times with consistent initial inputs, assessing output variability through LLM consensus rate across runs. The LLM consensus rate is calculated using the formula:

$$\text{LLM consensus rate} = \frac{\sum_{i=1}^n I(f_i = f_{\{majority\}})}{n}$$

Where:

n is the total number of runs; f_i is the generic frame assigned in run i ;

$f_{\{majority\}}$

is the most frequently assigned generic frame across all runs; $I()$ is an indicator function that equals 1 when the condition is true and 0 otherwise. This metric ranges from 0 (no consensus) to 1 (perfect consensus).

Furthermore, manual validation by three independent researchers quantified performance through frame consistency rate, ensuring robust, human-validated reliability (Zhou et al., 2024). This metric

Table 1
Vocabulary for S-WEAT.

Attribute	Vocabulary
Warm	friendly, well-intentioned, good-natured, agreeable, trustworthy, warm, sincere, nice, kind, dependable, supportive
Cold	unfriendly, hateful, untrustworthy, cold, mean, dishonest, vicious, deceitful, disloyal, selfish, hostile
Competent	competent, confident, capable, efficient, intelligent, skillful, skilled, qualified, proficient, able, smart
Incompetent	Incompetent, uncertain, ignorant, inefficient, unintelligent, unskilled, helpless, unqualified, dumb, foolish, inept

Model: o4-mini

Parameters: response_format={"type": "json_object"}, reasoning_effort="high", store=True

SYSTEM_PROMPT = ""

You are an expert in post-processing topic modeling results, specializing in analyzing viewer comments from the "Break 50" golf challenge featuring Bryson DeChambeau and Donald Trump—an event where the duo played 18 holes aiming for a score below 50 strokes. Your task involves analyzing BERTopic outputs and hierarchical clustering data to classify topics into coherent frames and recommend potential merges for improved clarity and thematic consistency.

INPUT UNDERSTANDING

You will analyze:

Topic Modeling Results

- Numeric topic IDs
- Topic labels with key terms
- Keyword representations
- GPT labels
- Representative documents

Hierarchical Clustering Results

- Lists of topic IDs
- Associated distance metrics

GENERIC FRAME ANALYSIS

Classify each topic into generic frames:

- Athletics
- Entertainment
- Politics/Controversy
- MANUAL (when classification is unclear)

MERGE SUGGESTION METHODOLOGY

For each topic:

1. Identify topics assigned to the same generic frame
2. Analyze their semantic similarity by comparing:
 - Keyword relevance in keyword representations
 - Thematic similarity in representative documents
 - Proximity in the hierarchical clustering results
3. Suggest merging topics with high similarity ONLY within the same generic frame
4. If unable to make a clear determination, assign "MANUAL"

OUTPUT FORMAT

Return a JSON object with the following structure:

```
{
  "frame_analysis": [
    {
      "generic_frame": "<generic_frame_name>",
      "topics": [
        { "topic_id": <int>, "topic_name": "<gpt_label>" }
      ],
      "total_topics": <int>
    }
  ],
  "merge_suggestions": [
    {
      "generic_frame": "<generic_frame_name>",
      "suggestion": [
        {
          "topic_ids": [<id1>, <id2>, "..."],
          "topic_name": "<gpt_label>",
          "rationale": "<merge_justification>"
        }
      ]
    }
  ],
  "summary": {
    "topic_counts": {
      "<generic_frame_name>": { "topic_ids": [<id1>, <id2>, "..."], "count": <int> },
      "MANUAL": { "topic_ids": [<id1>, <id2>, "..."], "count": <int> },
      "total_topics": <int>
    },
    "merge_suggestions": {
      "suggested_merges": {
        "frames": { "<generic_frame_name>": [<id1>, <id2>, "..."],
          "count": <int>
        },
        "manual_review": { "topic_ids": [<id1>, <id2>, "..."], "count": <int> }
      }
    }
  }
}
```

Fig. 2. o4-mini system prompt.

evaluates the degree of concordance between LLM classifications and human judgments.

$$\text{Frame consistency rate} = \frac{f_{LLM \text{ and Human}}}{f_{LLM \text{ or Human}}}$$

Where:

$f_{LLM \text{ and Human}}$ is the number of generic frames that both the LLM and human evaluators assign in agreement; $f_{LLM \text{ or Human}}$ represents the total number of generic frames assigned by either LLM or human evaluators. This metric ranges from 0 (no consistency) to 1 (perfect consistency).

3.3.4. LLM fine-tuning for enhanced automation and accuracy

To augment the accuracy, efficiency, and scalability of our analytical approach, we further employed fine-tuning techniques using GPT-4o-mini (OpenAI, 2025b), chosen explicitly for its optimal balance of high analytical performance, lower operational latency, and cost efficiency compared to more resource-intensive models such as GPT-4.1. The fine-tuning process began with BERTopic outputs, from which 50 comments per identified topic were randomly selected. As illustrated in Figs. 4 and 5, these documents were subsequently analyzed by GPT-4.1, utilizing an explicitly crafted system prompt structured for chain-of-thought (COT) reasoning and topic assignment (Wei et al., 2022). The structured JSONL outputs, consisting of COT rationales and topic classifications, were then split into training (960 entries, 80.00 %

and validation (240 entries, 20.00 %) datasets, with content verification conducted by three researchers to ensure accuracy, consistency, and theoretical alignment (Baier et al., 2025).

We fine-tuned GPT-4o-mini with standardized hyperparameters, three epochs, a batch size of one, and a learning rate multiplier of 1.80, while rigorously tracking training loss, validation loss, and full validation loss metrics to ensure optimal learning progression. This meticulous fine-tuning significantly improved the automation pipeline's classification accuracy and efficiency, facilitating robust, theoretically grounded analyses of our extensive dataset.

4. Results

4.1. Identification and interpretation of frames

4.1.1. Topic clusters and keyword representations

Topic modeling analysis was conducted with BERTopic, employing a cluster parameter set at 50. This value was selected to effectively balance interpretability and granularity in topic generation, ensuring meaningful thematic coherence without overly fragmenting or oversimplifying nuanced viewer discussions present within the dataset.

The initial modeling yielded 45 distinct clusters, including a substantial proportion of outliers ($n = 25,699$; approximately 54.32 % of the total dataset). The outliers, labeled as "-1", encompassed diverse

BERTopic results excerpt:

```
{
  "Topic": 1,
  "Name": "I_golf video_old_golf_best golf",
  "Representation": [
    "golf video",
    "old",
    "golf",
    "best golf",
    "youtube golf",
    "stroke",
    "watched",
    "putting",
    "youtube",
    "putting stroke"
  ],
  "GPT": [
    "Golf videos and discussions about older players"
  ],
  "Representative_Docs": [
    "Best golf video on YouTube",
    "Say whatever the heck you want but to be pushing and playing golf at that pace is pretty remarkable Hits them straight Pretty good distance He knows his clubs Hes confident in his hits Has a solid putting game Let me remind you hes years old And hes having a good time doing it Thats the most impressive thing of all",
    "Best YouTube golf video of all time"
  ]
},
...
```

Hierarchical clustering results excerpt:

```
{
  "Topics": [
    0,
    1,
    2,
    3,
    4
  ],
  "Distance": 0.815756223775752
},
{
  "Topics": [
    0,
    1,
    2,
    3
  ],
  "Distance": 0.8039110489427046
},
...
```

o4-mini output excerpt:

```
{
  "frame_analysis": [
    {
      "generic_frame": "Athletics",
      "topics": [
        {
          "topic_id": 0,
          "topic_name": "Bryson DeChambeau and Donald Trump Golf Collaboration"
        },
        {
          "topic_id": 1,
          "topic_name": "Golf videos and discussions about older players"
        }
      ],
      "merge_suggestions": [
        {
          "generic_frame": "Athletics",
          "suggestion": [
            {
              "topic_ids": [
                2,
                9,
                30
              ],
              "topic_name": "Donald Trump's golf performance and praise",
              "rationale": "These topics all focus on praising Trump's golf skills and discussing his overall performance, so merging them would streamline commentary on his athletic ability."
            }
          ]
        }
      ]
    }
  ],
  ...
}
```

Fig. 3. o4-mini input and output excerpts.

Model: gpt-4.1/4o-mini

Parameters: response_format={"type": "json_object"}, temperature=0, max_tokens=256

SYSTEM_PROMPT = ""

You are an expert content analyst of Break 50 video comments. Select the single best topic from the TOPICS list below. Perform the step-by-step analysis for each comment. Return a JSON object exactly following Output Format.

TOPICS

```
TOPICS = [
  { "topic_number": 0, "topic_name": "Donald Trump's golf skills and reputation" },
  { "topic_number": 1, "topic_name": "Positive reactions to DeChambeau and Trump collaboration" },
  { "topic_number": 2, "topic_name": "Video-hype superlatives" },
  { "topic_number": 3, "topic_name": "Broader political debate: media, policy & national issues" },
  { "topic_number": 4, "topic_name": "Trump-vs-Biden golf showdown fantasies" },
  { "topic_number": 5, "topic_name": "Media-distrust & TDS accusations" },
  { "topic_number": 6, "topic_name": "Trump's behavior driving golf carts" },
  { "topic_number": 7, "topic_name": "DeChambeau YouTube subscriber milestones and viral video growth" },
  { "topic_number": 8, "topic_name": "Assassination-attempt timing & ear-injury speculation" },
  { "topic_number": 9, "topic_name": "Trump's birdies/eagle highlights & quirky putting stroke" },
  { "topic_number": 10, "topic_name": "How to address former presidents" },
  { "topic_number": 11, "topic_name": "Pro-Trump enthusiasm, GOAT claims & voting pledges" },
  { "topic_number": 12, "topic_name": "Sexual innuendo & jokes" },
  { "topic_number": 13, "topic_name": "Celebrity guest wish-list" },
  { "topic_number": 14, "topic_name": "Kamala-Harris ads & follow-up commentary" },
  { "topic_number": 15, "topic_name": "Neville the caddy shout-outs & jokes" },
  { "topic_number": 16, "topic_name": "Repetition of the word partner" },
  { "topic_number": 17, "topic_name": "Debate over playing from red or women's tees" },
  { "topic_number": 18, "topic_name": "Comments on Trump's age & fitness" },
  { "topic_number": 19, "topic_name": "Trump cheating & integrity debates" },
  { "topic_number": 20, "topic_name": "Patrick Reed admiration & Trump connection" },
  { "topic_number": 21, "topic_name": "Wounded Warrior Project fundraising debate" },
  { "topic_number": 22, "topic_name": "Criticisms of golfing with Trump" },
  { "topic_number": 23, "topic_name": "Awkward social greetings" }
]
```

Analysis Process

1. Extract (max 25 words)

Consider implicit or contextual elements:

- Potential references to ongoing narratives or controversies
- Subtext behind brief or cryptic comments
- Possible sarcasm, humor, or figurative language
- Cultural or political references that may not be explicit

2. Evaluate (max 25 words)

Determine the primary focus and rating:

- Identify which topic number captures the central message or intent

- Assess relevance using these specific criteria:

- * Rating 5: Perfectly matches topic with clear, direct references
- * Rating 4: Strong match with minimal competing elements
- * Rating 3: Clear match but with significant competing elements
- * Rating 2: Partial match with dominant competing elements
- * Rating 1: Minimal connection

- rating and topic_number should be based on reasoning. If rating falls below 3, select "MANUAL" as the topic_number

Output Format

Return a JSON object exactly like:

```
JSON = [
  {
    "extract": "Contextual elements and implicit meaning (max 25 words)",
    "evaluate": "Determination of primary focus and matching strength based on rating scale criteria (max 25 words)",
    "rating": "Integer (1-5)",
    "topic_number": "Integer (0-23) or 'MANUAL'"
  }
]
```

Fig. 4. gpt-4.1/4o-mini system prompt.


```
{
  "extract": "Comment criticizes DeChambeau for associating
with Trump, referencing Trump's legal issues and moral
implications of the collaboration.",
  "evaluate": "Primary focus is criticism of golfing with Trump,
referencing his criminal status. Strong match for criticisms of
golfing with Trump, minimal competing elements.",
  "rating": 5,
  "topic_number": 22
}
```

Fig. 5. gpt-4.1/4o-mini output.

commentaries that did not fit neatly within identified clusters. Preliminary topic extraction revealed several salient themes, such as collaboration between Bryson DeChambeau and Donald Trump ($n = 4189$), discussions of golfing skills ($n = 2493$), broader political debates ($n = 1639$), positive viewer reactions and enthusiasm ($n = 1502$), and comparisons between Donald Trump and Joe Biden's golfing capabilities ($n = 1446$). Other major themes captured included negativities toward mainstream media, humorous observations, and speculation about specific events related to the featured individuals. BERTopic results and visualizations are provided in the online appendix.¹

4.1.2. Enhanced frame classification through human-AI collaboration

Despite the informative insights from these initial results, the high proportion of outliers and apparent semantic overlaps among certain clusters highlighted potential limitations in interpretability. To structurally assess thematic relationships and address the complexity inherent within the large dataset, hierarchical clustering analysis was subsequently conducted. Hierarchical clustering systematically organized topics based on semantic proximity, identifying clusters suitable for merging to enhance coherence and reduce redundancy (Grootendorst, 2022).

Leveraging OpenAI's o4-mini model, we input detailed hierarchical clustering results along with initial topic modeling data. The analysis was executed ten times with a consistent, customized system prompt specifically designed for hierarchical framing classification (Fig. 3). Despite the probabilistic nature of LLM outputs, our LLM assisted approach identified consistent generic frames across runs, yielding a high LLM consensus rate of 0.94 and a high frame consistency rate of 0.98. This strong agreement, both across multiple independent model runs and between LLM and human evaluators, demonstrates robustness in generic frame classification decisions. However, issue-specific frames required further review due to nuanced thematic distinctions. Consequently, manual assessments informed by the model's recommendations guided final merging decisions.

Following this collaborative human-AI analysis, we merged closely related topics to consolidate overlapping themes, aiming to enhance interpretability. For instance, initial separate topics related to Trump's golfing skills, viewer enthusiasm, media skepticism, and discussions of viral subscriber milestones were effectively merged into coherent clusters. The final merged topic set included clearly delineated themes such as Trump's Golf Skills and Reputation (combined topics 1, 2, 9, 30, 40), video-hype superlatives (topics 4, 23, 27, 36, 38), media distrust and Trump derangement syndrome (TDS) discussions (topics 6, 28), and subscriber milestones and viral growth (topics 8, 25, 42). Additional merges consolidated related topics such as humorous and risqué commentary, pro-Trump enthusiasm, and anti-Trump debates. As illustrated in Table 2, the finalized 24 coherent, semantically distinctive topics

Table 2

BERTopic results following hierarchical merging.

Topic	Name	Generic frame	Count
0	Trump's golf skills and reputation	Athletics	5917
1	Positive reactions to DeChambeau and Trump collaboration	Entertainment	4189
2	Video-hype superlatives	Entertainment	1832
3	Broader political debate: media, policy & national issues	Politics/ Controversy	1639
4	Trump-vs-Biden golf showdown fantasies	Athletics	1446
5	Media-distrust & TDS discussions	Politics/ Controversy	1185
6	Trump's behavior driving golf carts	Entertainment	856
7	DeChambeau YouTube subscriber milestones and viral video growth	Entertainment	717
8	Assassination-attempt timing & ear-injury speculation	Politics/ Controversy	513
9	Trump's birdies/eagle highlights & quirky putting stroke	Athletics	465
10	How to address former presidents	Politics/ Controversy	419
11	Pro-Trump enthusiasm, GOAT claims & voting pledges	Politics/ Controversy	375
12	Sexual innuendo & jokes	Entertainment	340
13	Celebrity guest wish-list	Entertainment	329
14	Kamala-Harris ads & follow-up commentary	Politics/ Controversy	252
15	Neville the caddy shout-outs & jokes	Entertainment	247
16	Repetition of the word partner	Entertainment	182
17	Debate over playing from red or women's tees	Athletics	150
18	Comments on Trump's age & fitness	Athletics	146
19	Trump cheating & integrity debates	Politics/ Controversy	126
20	Patrick Reed admiration & Trump connection	Entertainment	79
21	Wounded Warrior Project fundraising debate	Politics/ Controversy	75
22	Criticisms of golfing with Trump	Politics/ Controversy	75
23	Awkward social greetings	Entertainment	56

Note: The online repository contains the complete BERTopic results (topic keywords, KeyBERT, MMR, and GPT representations) and all visualizations (keyword scores, intertopic distance map, hierarchical clustering, topic heatmap).

significantly reduced the number of fragmented or redundant clusters, aside from remaining outliers, thus providing an analytically robust framework for subsequent detailed framing and thematic interpretation.

4.2. Performance of fine-tuned GPT-4o-mini in automated classification

The fine-tuned GPT-4o-mini model² demonstrated effective learning, with training loss decreasing from 2.20 to 0.15 and accuracy improving from 0.65 to 0.90, while exhibiting minimal overfitting and strong generalization capabilities. Detailed fine-tuning procedures and metrics are available in the online appendix. The model then classified 47,309 viewer-generated comments into 24 coherent issue-specific topics previously identified (Table 2).

This classification process effectively addressed RQ1. The most prevalent topics included Trump's golf skills and reputation ($n = 6657$; 14.07 %), positive reactions to DeChambeau and Trump collaboration ($n = 5783$; 12.22 %), and broader political debate: media, policy, and nationalism ($n = 5621$; 11.88 %). Additional prominent topics were video-hype superlatives ($n = 4254$; 8.99 %) and pro-Trump enthusiasm, GOAT claims, and voting pledges ($n = 3225$; 6.82 %). A smaller proportion of comments were categorized under narrower themes, such as media-distrust and TDS discussions ($n = 2349$; 4.97 %), Trump-vs-Biden

¹ The online appendix and supplemental files are available at <https://github.com/TyrealQ/Break50>.

² The model ID for our fine-tuned GPT-4o-mini is ft:gpt-4o-mini-2024-07-18:qian:b50:BUOe6orE.

golf showdown fantasies ($n = 2299$; 4.86 %), and DeChambeau YouTube subscriber milestones and viral video growth ($n = 1611$; 3.41 %). Comments that did not align clearly with these established topics, often due to ambiguity or lack of coherent meaning (e.g., “Just another soy boy”), were grouped under the category labeled Manual ($n = 2222$; 4.70 %), reflecting their inability to fit neatly into the refined topic structure. The overall topic distribution confirms the efficacy of the GPT-4o-mini fine-tuning approach in achieving nuanced and precise automated topic classification.

4.3. S-WEAT analysis of generic and issue-specific frames across platforms

In response to RQ2, semantic evaluations of warmth and competence were conducted in alignment with SCM and operationalized through S-WEAT. The detailed results are presented in Tables 3 and 4, with visual distributions in Fig. 6 for issue-specific frames. Validation against a stratified subsample of human-coded comments demonstrated strong convergence ($r = 0.85$), indicating that the embedding-based measures captured audience perceptions in line with SCM’s conceptual dimensions. To further test robustness, we compared results from the text-embedding-3-small model with an alternative embedding model (BAAI/bge-m3). Despite substantial differences in architecture and training data, the S-WEAT scores showed a high degree of alignment (warmth-cold: $r = 0.79$; competence-incompetence: $r = 0.78$), providing evidence of convergent validity and reinforcing that our operationalization of warmth and competence is stable across models rather than tied to a single embedding framework.

Across the three generic frames, entertainment consistently produced strongly positive warmth and competence evaluations across platforms, while politics/controversy exhibited the widest variability, including multiple warmth-competence sign reversals. These fluctuations were most apparent in morally and politically contentious topics, reflecting the competitive frame environment predicted by framing theory (De Vreese, 2005). Athletics discourse tended to produce moderate to high competence and moderate warmth, but several subtopics showed platform-driven divergences, especially in competence assessments tied to age and political imagination scenarios.

As to RQ3, we focus on platform-specific divergences and theoretical outliers to illustrate how context shapes evaluations. Entertainment was the most frequent generic frame ($n = 17,242$; 38.24 %), with high mean warmth (0.74) and competence (0.74) across platforms (YouTube: warmth = 0.72, competence = 0.78; Instagram: warmth = 0.75, competence = 0.74). Within this frame, positive reactions to DeChambeau-Trump collaboration (Topic 1; $n = 5783$) reached the highest warmth (0.97) and competence (0.93) of any topic in the dataset, reflecting a high-warmth/high-competence “admiration” quadrant profile in SCM. This aligns with framing theory’s expectation that entertainment-focused portrayals can foreground non-threatening, skill-oriented attributes, fostering both affective and competence-based positivity.

Video-hype superlatives (Topic 2; $n = 4254$) demonstrated similarly high evaluations (warmth = 0.78; competence = 0.85), with Instagram showing greater proportional engagement (17.70 %) than YouTube (7.59 %). While most entertainment issue-specific frames aligned with this positive pattern, some revealed theoretically interesting deviations. For instance, Patrick Reed admiration and Trump connection (Topic 20;

$n = 117$) was a low-frequency but high-variance case: YouTube comments scored high on competence (1.21) and warmth (0.74), whereas Instagram comments were near neutral (warmth = 0.01, competence = 0.26), illustrating how platform context can attenuate admiration despite similar thematic framing. Another outlier, awkward social greetings (Topic 23; $n = 736$), showed moderate warmth (YouTube = 0.27; Instagram = 0.43) but negative competence (YouTube = -0.28 ; Instagram = -0.10), producing a “warm-but-incompetent” profile. Comments like “49:02 he tried to fist bump but Trump said nah lmao” and “This is like when you have to hang out with your grandpa and kinda talk to him like he’s a kid” illustrate the affectionate but ability-doubting tone reflected in the scores.

Politics/controversy ($n = 15,450$; 34.27 %) displayed near-neutral mean warmth (0.09) and slightly negative competence (-0.05), though the variance across topics was substantial. The dominant issue-specific frame, broader political debate: media, policy, and nationalism (Topic 3; $n = 5621$), consistently occupied SCM’s contempt quadrant (warmth = -0.17 , competence = -0.32) across platforms, as did media-distrust and TDS discussions (Topic 5) and criticisms of golfing with Trump (Topic 22), with warmth ranging from -0.18 to -0.43 and competence from -0.34 to -0.57 . Illustrative comments include “All Trump does is name call, insult, and act like a child” and “How dare you use this platform to promote a convicted sexual predator”, which combine affective rejection with perceived moral and capability failings.

However, notable counter-patterns emerged. Pro-Trump enthusiasm, GOAT claims, and voting pledges (Topic 11; $n = 3225$) scored high on both warmth (0.82) and competence (0.74), showing that even within a contentious frame category, politically aligned discourse can generate admiration-type evaluations. Two issue-specific frames showed clear sign reversals between platforms: Kamala-Harris ads and follow-up commentary (Topic 14) shifted from positive warmth on YouTube (0.34) to slightly negative warmth on Instagram (-0.02), despite negative competence across both. For Trump cheating and integrity debates (Topic 19), competence shifted from mildly positive on YouTube (0.14) to negative on Instagram (-0.14) while warmth remained negative across platforms. YouTube discussion often coupled grudging skill recognition with skepticism, for example, “Trump is clearly an accomplished golfer. Honest question though, was he given any second chances on ANY shot and it was edited together?”, whereas Instagram comments framed the same behavior as categorical evidence of inability or dishonesty, for example, “Trump has the greatest foot-wedge of all time!! #liarandacheat” and “Trump cheats lol oh everyone knows that.” Such reversals suggest that the same thematic content may activate different interpretive schemas depending on platform characteristics and community composition.

Athletics ($n = 12,394$; 27.49 %) averaged moderately positive warmth (0.31) and notably positive competence (0.49). The most prevalent issue-specific frame, Trump’s golf skills and reputation (Topic 0; $n = 6657$), reinforced this profile (warmth = 0.44, competence = 0.73), with greater salience on YouTube (15.08 %) than Instagram (7.84 %). This finding underscores the role of sports performance framing in fostering skill-based respect even amid politically charged contexts. Trump-vs-Biden golf showdown fantasies (Topic 4) displayed low warmth overall (-0.05) but positive competence (0.12), representing an “envy” quadrant profile where ability is acknowledged without interpersonal warmth. This is evident in comments such as “Trump is a legit golfer. I can’t believe Biden even pretended he could break 100”. Comments

Table 3
Generic frame and S-WEAT analysis results.

Generic frame	Topic	Count	Share	YT share	INS share	WC	CI	YTCW	YTCI	INSWC	INSCI
Entertainment	1, 2, 6, 7, 12, 13, 15, 16, 20, 23	17242	38.24	36.09	51.84	0.74	0.74	0.72	0.78	0.75	0.74
Politics/controversy	3, 5, 8, 10, 11, 14, 19, 21, 22	15450	34.27	34.87	30.45	0.09	-0.05	0.08	-0.02	0.09	-0.05
Athletics	0, 4, 9, 17, 18	12394	27.49	29.04	17.71	0.31	0.49	0.31	0.48	0.31	0.49

Note: YT = YouTube, INS = Instagram, WC = warmth-cold score, CI = competence-incompetence score.

Table 4

Issue-specific frame and S-WEAT analysis results.

Topic	Issue-specific frame	Count	YTShare	INSShare	WC	CI	YTWc	YTcI	INSWC	INSCI
0	Trump's golf skills and reputation	6657	15.08	7.84	0.44	0.73	0.36	0.66	0.45	0.73
1	Positive reactions to DeChambeau and Trump collaboration	5783	11.86	14.47	0.97	0.93	0.87	0.87	1.00	0.94
3	Broader political debate: media, policy & national issues	5621	11.91	11.69	-0.17	-0.32	-0.15	-0.31	-0.17	-0.32
2	Video-hype superlatives	4254	7.59	17.70	0.78	0.85	0.67	0.87	0.82	0.84
11	Pro-Trump enthusiasm, GOAT claims & voting pledges	3225	7.12	4.94	0.82	0.74	0.85	0.95	0.81	0.72
5	Media-distrust & TDS discussions	2349	5.26	3.16	-0.18	-0.34	-0.10	-0.31	-0.19	-0.34
4	Trump-vs-Biden golf showdown fantasies	2299	4.96	4.23	-0.05	0.12	0.25	0.37	-0.09	0.08
Manual	Manual	2222	4.42	6.43	0.37	0.06	0.37	0.03	0.37	0.07
7	DeChambeau YouTube subscriber milestones and viral video growth	1611	3.70	1.57	0.59	0.86	0.25	0.55	0.62	0.88
18	Comments on Trump's age & fitness	1568	3.55	1.85	0.41	0.06	0.27	-0.14	0.42	0.08
6	Trump's behavior driving golf carts	1561	3.66	1.05	0.35	0.41	0.23	0.19	0.36	0.42
9	Trump's birdies/eagle highlights & quirky putting stroke	1329	2.93	2.05	0.19	0.46	0.23	0.50	0.18	0.45
13	Celebrity guest wish-list	1214	1.58	8.68	0.75	0.79	0.84	0.91	0.67	0.68
12	Sexual innuendo & jokes	1034	2.20	2.10	0.45	0.29	0.51	0.28	0.44	0.29
22	Criticisms of golfing with Trump	945	1.85	2.93	-0.43	-0.57	-0.35	-0.57	-0.45	-0.57
8	Assassination-attempt timing & ear-injury speculation	936	1.84	2.81	0.42	0.16	0.47	0.25	0.41	0.14
10	How to address former presidents	828	1.92	0.70	0.22	-0.16	0.26	-0.07	0.22	-0.17
23	Awkward social greetings	736	1.48	2.05	0.40	-0.13	0.27	-0.28	0.43	-0.10
19	Trump cheating & integrity debates	694	1.47	1.47	-0.45	-0.10	-0.34	0.14	-0.47	-0.14
17	Debate over playing from red or women's tees	541	1.23	0.59	0.24	0.36	0.55	0.57	0.22	0.34
15	Neville the caddy shout-outs & jokes	534	1.25	0.35	0.63	0.55	0.81	0.88	0.62	0.54
14	Kamala-Harris ads & follow-up commentary	515	1.21	0.36	0.00	-0.13	0.34	0.25	-0.02	-0.15
16	Repetition of the word partner	398	0.90	0.50	0.95	0.87	0.96	0.87	0.95	0.87
21	Wounded Warrior Project fundraising debate	337	0.76	0.41	0.61	0.31	0.44	0.16	0.62	0.32
20	Patrick Reed admiration & Trump connection	117	0.28	0.05	0.02	0.29	0.74	1.21	0.01	0.26

Note: YT = YouTube, INS = Instagram, WC = warmth-cold score, CI = competence-incompetence score.

on Trump's age and fitness (Topic 18) were uniformly warm (0.41) but diverged on competence, from slightly negative on YouTube (-0.14) to slightly positive on Instagram (0.08). Remarks from YouTube like *"I feel very sorry for him. He deserves to spend his last years with his family"* exemplify this pity-quadrant mix of sympathy and diminished perceived ability, indicating the potential for benevolent stereotyping when age or health is foregrounded.

Figs. 7 and 8 present the analysis of generic and issue-specific frames from July 23 to August 11, 2023, respectively. On the video's YouTube release date (July 23), entertainment frame generated the highest cross-platform discussion, with athletics and politics/controversy following closely behind. By July 25, comment frequencies decreased markedly across all three generic frames. Nevertheless, the politics/controversy frame experienced a proportionally smaller decline than entertainment and athletics, sustaining a more consistent presence throughout the observation period. A similar pattern emerged for issue-specific frames, where broader political debate: media, policy, and nationalism became the most discussed frame on YouTube on July 26. Instagram, however, exhibited a distinctive pattern, wherein two separate posting events, one on 23 July and another on 25 July, triggered distinct peaks of frames of video-hype superlatives and DeChambeau YouTube subscriber milestones and viral video growth. Collectively, this nuanced interplay between content and platform contexts underscores how audience evaluations are dynamically shaped by both discourse framing and the distinct user communities on YouTube and Instagram.

5. Discussion

In this study, we analyzed social media reactions to Donald Trump's appearance on Break 50, uncovering a diverse interpretive landscape shaped by generic and issue-specific frames. In doing so, our study bridges and extends framing theory and SCM by demonstrating how these frameworks can be operationalized to analyze large-scale UGC. While entertainment framing elicited high warmth and competence perceptions, politics/controversy invoked predominantly negative evaluations, and athletics framing centered around Trump's perceived competence. Issue-specific frames further nuanced these perceptions, illustrating cross-platform divergences between YouTube and Instagram. By illustrating how sports-politics intersections prompt

individuals to draw on cognitive schemas from multiple domains, our findings illuminate the complex interplay between frame setting, political identity, and character judgment.

5.1. Theoretical implications

The current study contributes to the application of framing theory and SCM within UGC contexts. Framing theory traditionally emphasizes how media entities strategically shape audience interpretations through selective emphasis of information (Guo et al., 2021; Jiang et al., 2024; Tan and Cheng, 2024). Our findings extend this perspective by illustrating how audiences, through interactive dialogue, actively construct, adapt, and contest frames in ways that both align with and complicate prior expectations (Ocal and Crowston, 2024; Xu et al., 2024). This dynamic is evident in the distinction between generic and issue-specific frames. While generic frames yielded interpretive tendencies consistent with prior research, their issue-specific counterparts often revealed deviations that complicate these established associations. For instance, politics/controversy frames not only reinforced ideological divides, as anticipated, but also generated unexpected moments of humor, admiration, or even reluctant respect (e.g., assassination-attempt timing & ear-injury speculation), suggesting that conflict-oriented frames may carry ambivalent valences in participatory settings. These patterns underscore that in digital environments, framing is not simply the reproduction of stable, generic categories but an iterative process where issue-specific expressions can subvert or reconfigure expected meanings. Thus, our study advances framing theory by demonstrating that audience-driven framing in UGC is inherently flexible, contextually contingent, and capable of reshaping the interpretive logics embedded in generic frames.

Further, our study extends the theoretical scope of SCM by demonstrating its explanatory power in the unstructured and dynamic realm of UGC (Liu et al., 2024). Unlike the highly controlled laboratory or survey-based settings in which SCM has been predominantly examined (Bennett et al., 2019; Funk, 1997; Kervyn et al., 2012, 2022), our study draws on spontaneous, large-scale social media interactions, situating SCM within the naturally occurring meaning-making practices of online communities. These organically produced comments offer a dynamic, context-rich lens into how individuals interpret and evaluate public

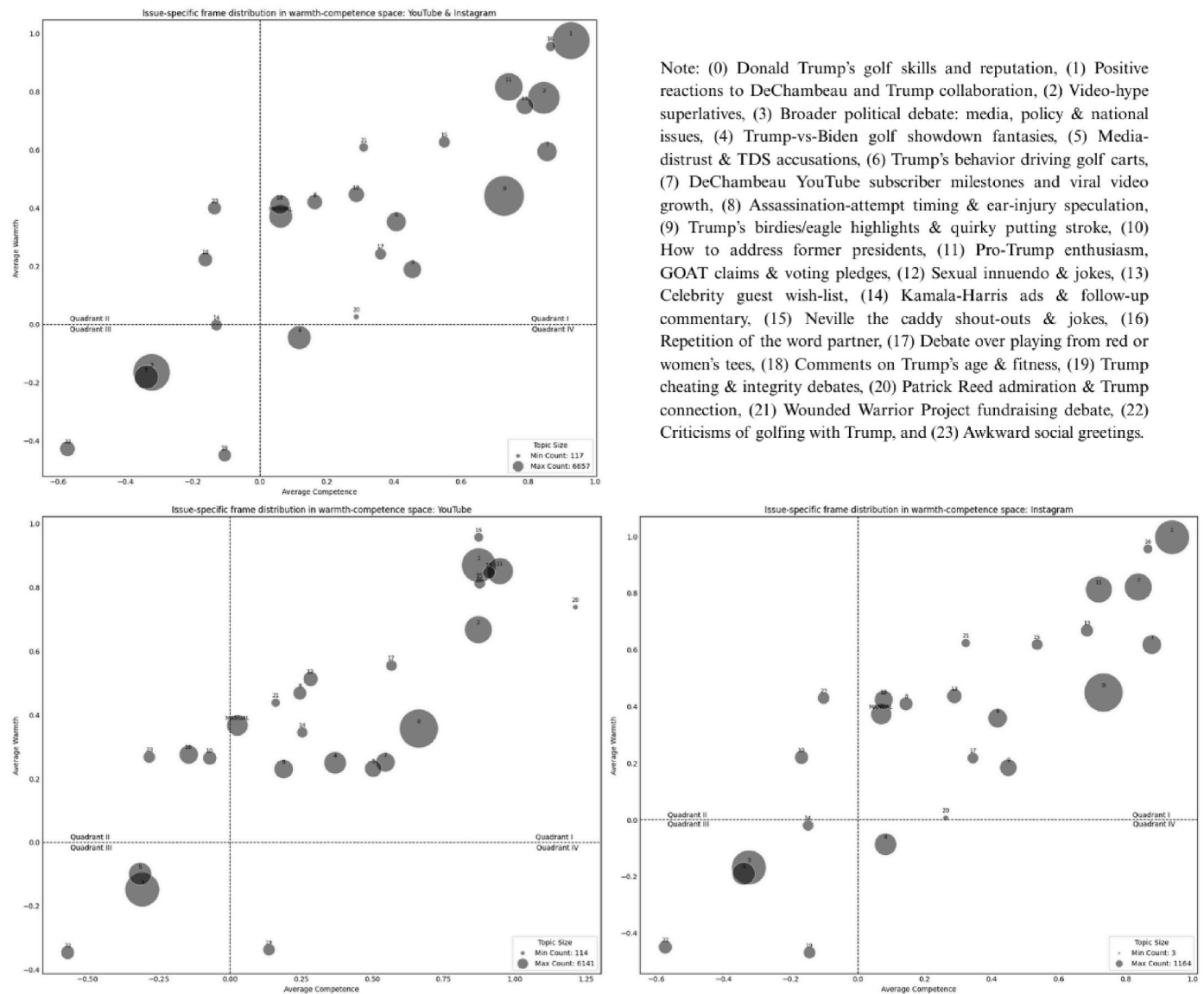


Fig. 6. Issue-specific frame distributions in warmth-competence space across platforms.

Note: Quadrants I-IV represent distinct stereotype patterns: admired groups (high warmth, high competence), pitied groups (high warmth, low competence), contemptuous groups (low warmth, low competence), and envied groups (low warmth, high competence).

figures such as Trump and DeChambeau in cross-domain collaborations.

Importantly, our findings highlight not only the persistence of warmth and competence as distinct evaluative dimensions, but also their potential to combine in unexpected ways that refine SCM's theoretical assumptions. The emergence of warm-but-incompetent and cold-but-competent evaluations illustrates that audience perceptions of public figures do not always map neatly onto the canonical quadrants. For example, Trump was at times portrayed as warm-but-incompetent—emphasized through humorous depictions of his social interactions—while in other cases he was framed as cold-but-competent, admired for his golfing skills yet criticized for his political behavior. These patterns suggest that in participatory media environments, warmth and competence may operate as fluid dimensions, allowing users to simultaneously assign contradictory traits depending on the framing context. These observations indicate that stereotypes are not merely applied to public figures but actively reinterpreted, stretched, and contested through user discourse.

Lastly, the integration of SCM with framing theory clarifies how platform affordances condition these evaluative patterns. YouTube's long-form, video-centered environment allowed users to make direct

competence judgments based on visible evidence of Trump's physical performance, which sometimes diminished his competence ratings. By contrast, Instagram's image curation and textual brevity encouraged more favorable competence attributions in the absence of explicit negative visual cues. These platform-specific differences underscore how frames and modalities channel the expression of warmth and competence, suggesting that digital environments can amplify or suppress particular stereotype dimensions (A. Kirilenko et al., 2024; Peng, 2021). Taken together, these insights advance SCM by showing that warmth and competence operate as dynamic, context-sensitive constructs shaped by the interaction of framing processes and platform affordances.

5.2. Methodological implications

Our study demonstrates methodological innovations through an innovative human-AI collaboration pipeline integrating traditional NLP, reasoning-enhanced LLMs, LLM fine-tuning, and embedding-based semantic analysis. Initially, topic modeling via BERTopic efficiently identified overarching themes, showcasing substantial performance



Fig. 7. Generic frame temporal dynamics.

Note: Temporal distribution of generic frames (YouTube vs. Instagram, July 23–August 11, 2024), showing immediate entertainment and athletics spikes after Break 50's release, with politics/controversy sustaining longer-term visibility.

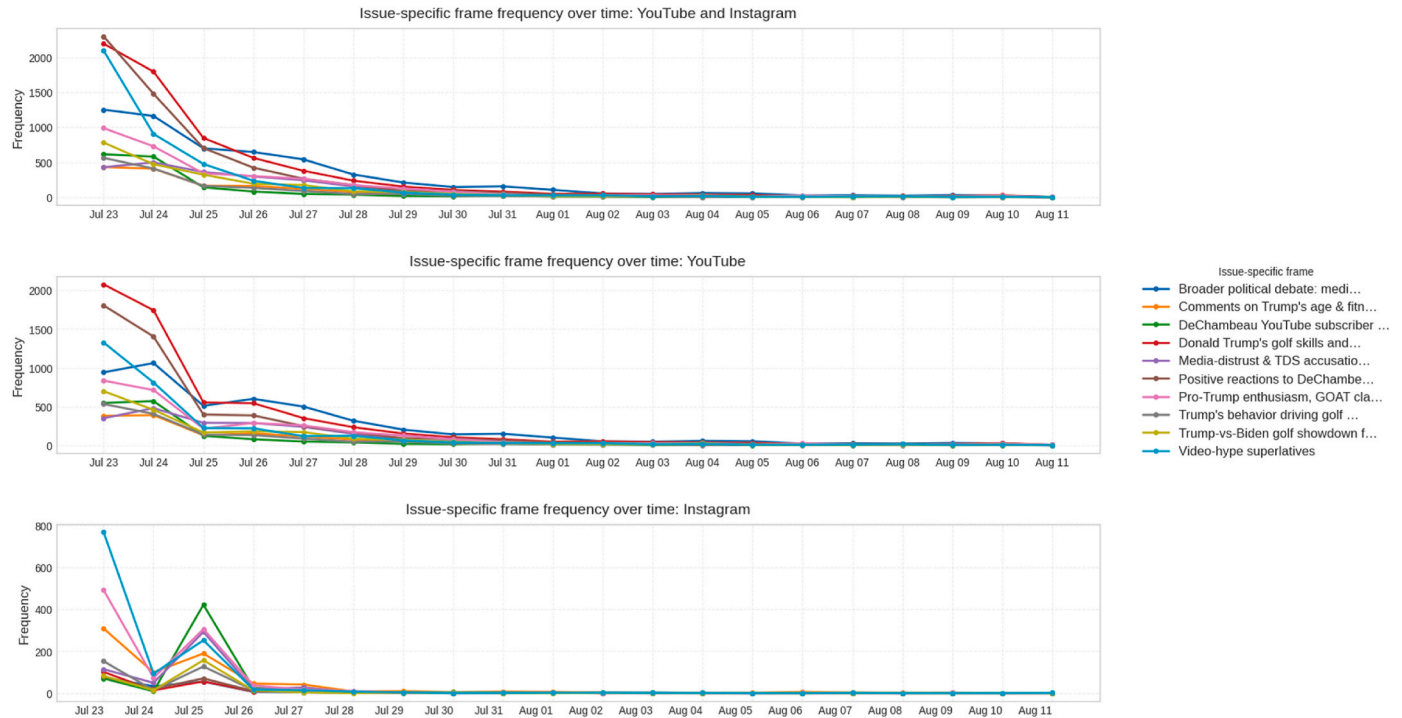


Fig. 8. Issue-specific frame (top 10 by frequency) temporal dynamics.

Note: Temporal distribution of issue-specific frames (YouTube vs. Instagram, July 23–August 11, 2024), illustrating platform-specific frame peaks from secondary Instagram posts and the prolonged presence of political debate on YouTube.

improvements by leveraging semantic embeddings for coherent clustering (Jiang et al., 2024; A. P. Kirilenko and Stepchenkova, 2025; T. Y. Qian et al., 2024). However, the inherent semantic overlaps and interpretive ambiguity necessitated advanced analytical refinement. Addressing this challenge, we incorporated a reasoning-oriented LLM (o4-mini) specifically prompted to interpret hierarchical clustering outputs and provide structured suggestions for merging semantically related topics. With human in the loop, this critical AI-driven reasoning step offered scalable interpretive guidance, enhancing thematic coherence considerably beyond manual clustering efforts (Berente et al., 2019).

Subsequently, fine-tuning GPT-4o-mini on the refined outputs of the reasoning-enhanced clustering process allowed precise and nuanced classification of tens of thousands of comments into meaningful issue-specific frames. This automated classifier provided transparent, auditable output, enhancing reliability and reducing the demand for extensive manual reconciliation (De Freitas et al., 2025). Lastly, the implementation of S-WEAT embedding-based analysis provided a robust quantification of warmth and competence evaluations within the classified comments. This embedding approach captured subtle semantic associations, extending analytical depth beyond traditional sentiment analyses and validating SCM dimensions empirically at scale.

Overall, our human-AI collaboration pipeline highlights the power of computational methods and the importance of human augmentation, offering researchers an efficient, replicable, and scalable workflow applicable beyond this specific context. By effectively integrating unsupervised topic modeling, reasoning-enhanced frame identification, fine-tuned LLM classification, and embedding-based semantic metrics, this methodological framework represents a meaningful advancement in computational social science, suitable for diverse content analysis tasks and platforms (N. Arora et al., 2024; Rossi et al., 2024).

5.3. Empirical implications

Our study offers guidance for managing partnerships and audience engagement strategies that extend beyond this empirical case and can inform decisions in partnerships that involve politically salient figures. Entertainment-related narratives proved particularly powerful for shaping perceptions. On both YouTube and Instagram, cooperative, non-threatening performance stories consistently reinforced positive evaluations of warmth and competence. Yet even within this favorable framing, unexpected moments, such as Trump's failed fist bumps and other socially awkward gestures (Topic 23), shifted him into a warm-but-incompetent quadrant. Instead of harming his image, these gaffes made him appear more relatable and authentic (Bennett et al., 2019; Kervyn et al., 2022). Strategically, humorous or self-deprecating appearances in sports-entertainment contexts can be intentionally deployed to soften partisan hostility, but this tactic requires framing control. A minor gaffe on the golf course may enhance relatability, whereas the same awkwardness reframed in political or leadership contexts could undercut perceptions of competence. Selective integration of "relatable gaffes" is therefore most effective in settings where competence is not the core evaluative criterion.

The persistence of contempt-quadrant placements in broader political debate (Topic 3) and media-distrust (Topic 5) shows that political contestation is "sticky" and difficult to neutralize once activated. Here, prosocial initiatives or credible third-party endorsements might be leveraged to re-anchor discourse toward more favorable evaluations. For example, charity-related mentions (e.g., Wounded Warrior Project fundraising, Topic 21) tended to produce higher competence ratings, even when embedded within political controversy. The key is to shift the narrative focus from partisan identity to shared values, allowing warmth and competence to be reframed through collective action rather than political alignment.

Indeed, frame-specific differences shape how risks and opportunities materialize. For example, competence deficits interpreted through

entertainment frames carried minor reputational cost, but when similar assessments appeared within politics/controversy frames, such as cheating debates (Topic 19), they signaled credibility concerns that could erode trust. These contrasts show that reputational thresholds are not fixed numerical cutoffs, but situational judgments shaped by framing. In practice, a polarizing figure can be an asset in one setting and a liability in another. Within entertainment frames, associations may enhance warmth and engagement in lighthearted campaigns; within politics/controversy frames, the same figure may undermine brand credibility if integrity and competence are core to audience expectations. Similarly, brands may find value in leaning into controversy when the goal is to activate strong in-group loyalty, for example, mobilizing partisan enthusiasm in narrowly targeted campaigns, but should avoid such associations when broad market appeal is the objective.

Platform environments further magnify these distinctions. On YouTube, long-form and curated content allows performance skill and cooperative play to be spotlighted, producing relatively positive competence signals even in contested areas. Instagram's meme-driven and highly visual culture, by contrast, often amplifies critical or mocking interpretations from the same content (Peng, 2021). Social media managers should therefore adapt campaigns to platform norms: use YouTube to showcase technical expertise and narrative depth, while leveraging Instagram for relational, aesthetic, and community-focused engagement (S. Qian et al., 2024).

Timing also plays a critical role. Break 50's July 23, 2024 release triggered immediate spikes in entertainment and athletics frames, while political controversy persisted well into August and became intertwined with the presidential election cycle. Strategists should anticipate that sports-entertainment appearances may be folded into larger political narratives. Pre-release "frame inoculation," i.e., priming audiences with collaborative or performance-based stories, can buffer against politicization, while staggered cross-platform releases can extend positive coverage windows and manage the slower decay of controversy.

Taken together, our findings underscore that there is no one-size-fits-all formula for navigating partnerships with polarizing figures. Instead, practitioners should treat warmth and competence evaluations as contextual signals interpreted through framing dynamics. The integrated framework developed here offers tools for identifying when moments of humor and relatability can be leveraged, when controversy might be selectively embraced for targeted audiences, and when avoidance or reframing is necessary to protect broad appeal.

5.4. Limitations and future research

While our study provides valuable insights into audience framing and trait evaluations in politically infused sports contexts, several limitations warrant careful consideration. First, we focused exclusively on textual comments from YouTube and Instagram, potentially overlooking critical interpretive nuances from audio and visual elements that may shape audience perceptions and framing processes. Future research should analyze user-generated images and videos to capture the complexity and richness that text alone cannot convey (A. Arora et al., 2025; Peng, 2021). Additionally, extending the analysis to other social media platforms (e.g., Twitter, TikTok, Reddit), each characterized by unique community norms and interaction patterns, would further enhance the comprehensive understanding of cross-platform audience dynamics (S. Qian et al., 2024).

Second, although framing theory and the SCM were effectively integrated, our interpretations relied predominantly on aggregated semantic evaluations. Individual differences, including psychological traits, political ideologies, and identity salience, were not explicitly measured. Future research should incorporate individual-level analyses—such as interviews, surveys, or experiments—to better elucidate cognitive mechanisms underlying audience interpretations and the nuanced application of framing and SCM within digital contexts (Kurdi et al., 2019; Larkin, 2025). Additionally, incorporating aspect-based

sentiment models, may help uncover how topic-specific evaluations vary by audience subgroup and issue salience (Perazzoli et al., 2022; T. Y. Qian et al., 2025; Zha et al., 2023), offering a more granular understanding of how meaning is co-created in online discourse.

Third, despite harnessing robust AI-human collaboration and fine-tuning, automated methods inherently risk overlooking subtle interpretive nuances like cultural idioms, sarcasm, or context-specific language uses. Moreover, pre-trained embeddings may contain biases that originate from their underlying language models, which often unconsciously incorporate societal stereotypes and unintended semantic associations from their training data (Bai et al., 2025). Future research could mitigate these concerns through explicit and implicit bias assessments, comparative validation against extensive human-coded datasets, or qualitative analyses to ensure interpretive accuracy and fairness (Yoganarasimhan and Iakovetskaia, 2024).

Finally, our dataset focused solely on reactions to a singular event involving a highly polarizing political figure, limiting generalizability. Further studies should examine a wider range of events, diverse sports, and varying political personas to determine the robustness of the interpretive frameworks and trait evaluations identified. Comparative analyses across multiple scenarios will significantly enhance our understanding of audience engagement dynamics at the intersection of politics, sports entertainment, and digital discourse.

6. Conclusion

In an era where political actors engage in brand-like behavior and lifestyle influencers wade into polarizing discourse, the boundaries between commerce, ideology, and entertainment are increasingly blurred. In the current study, we offer a theoretical and methodological blueprint for analyzing such entanglements, foregrounding audience agency while maintaining analytical rigor through computational tools. Our findings demonstrate that audience reactions at the intersection of sports, politics, and entertainment are not monolithic but are instead shaped by a diverse set of interpretive frames. Moving beyond a simplistic pro-versus-anti binary, our analysis reveals a complex landscape where audiences actively deploy varied interpretive frames. The interplay between perceived warmth and competence across these frames suggests a reconfiguration of what authenticity and expertise signify in hybrid celebrity-political performances.

The scope of these findings, however, is defined by clear boundaries in both our research design and case selection. Our reliance on textual data provides an incomplete picture that future multimodal analysis could substantially enrich. Furthermore, the focus on a single, polarizing individual tempers the generalizability of our findings. These limitations highlight the need to apply this analytical framework across less polarizing subjects and diverse cultural contexts to test its broader applicability. Even so, our research contributes to an emerging interdisciplinary conversation about the affective politics of engagement in digitally mediated service environments. For scholars and practitioners alike, the results suggest that engagement is no longer a proxy for approval or alignment but a site of symbolic contestation. Understanding this distinction is essential for interpreting consumer behavior in an attention economy shaped as much by emotional resonance as by ideological friction.

CRedit authorship contribution statement

Tyreal Yizhou Qian: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hua Gong:** Writing – review & editing, Visualization, Methodology, Investigation. **Chenglong Xu:** Writing – review & editing, Validation, Methodology, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT and Claude in order to improve the language and readability of the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jretconser.2025.104520>.

Data availability

The online appendix and supplemental files are available at <https://github.com/TyrealQ/Break50>.

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